Predicting Diabetes: Using Machine Learning in Healthcare to Make Americans Healthier

Michelle Kouba

Department of Data Science, Bellevue University

DSC 630: Predictive Analytics

Professor Hua

May 13, 2023

Milestone 4

Data Preparation

Given that this was a public database with no missing values and already mostly cleaned, the majority of data preparation steps didn't need to be performed. In addition, all data preparation was done in Milestone 3 and detailed in that section of this report. Data preparation steps in Milestone 3 include:

- Converted Boolean yes/no variables to 0 (No) and 1 (Yes).
- Removed Polydipsia due to its high correlation with Polyuria.
- Split data into a training and test set.
- Scaled the data using a MinMax scaler.

I used a pipeline with a grid search to test the accuracy of different settings/parameters for three different type of analyses to see which fit the data the best: random forest analyses, K-Nearest Neighbors, and logistic regression, for a total of 210 different combinations of these analyses. When I queried the results, the model of best fit was a random forest analyses, with a max setting of three features and 1,000 estimations for the optimal settings.

```
# Reading the "Early Stage Diabetes Risk Prediction Dataset" saved to the working directory.
df = pd.read_csv("Diabetes_Bangladesh.csv")
```

```
df.head()
```

	Age	Gender	Polyuria	Polydipsia	sudden weight loss	weakness	Polyphagia	Genital thrush	visual blurring	Itching	Irritability	delayed healing	partial paresis	muscle stiffness	Alopecia	Obesity	class
0	40	1	0	1	0	1	0	0	0	1	0	1	0	1	1	1	1
1	58	1	0	0	0	1	0	0	1	0	0	0	1	0	1	0	1
2	41	1	1	0	0	1	1	0	0	1	0	1	0	1	1	0	1
3	45	1	0	0	1	1	1	1	0	1	0	1	0	0	0	0	1
4	60	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1

```
# Creating features.
# Dropping Polydipsia because it's highly correlated with Polyuria.
X = df.drop(['class', 'Polydipsia'],axis=1)
y = df['class']
```

```
# Splitting the dataset into the training set and the test set.
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3, random_state = 0)
```

```
# Creating a pipline for the data to scale the data (MinMaxScaler) and run multiple
# Analyses and fit the best model to the data (multiple settings for each analyses).
# The analyses I will be trying to fit are logistic regression, KNN analyses, and a
# random forest analyses.
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import MinMaxScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
# Creating the MinMaxScaler
min_max_scaler = MinMaxScaler()
# Creating the search space for the analyses specifying these above analyses and
# settings.
search_space = [{'classifier': [LogisticRegression(solver='lbfgs', max_iter=5000)],
                 'classifier__penalty': ['l2'],
                 'classifier__C': np.logspace(0,4,10)},
                {'classifier': [RandomForestClassifier()],
                 'classifier__n_estimators': [10, 100, 1000],
                 'classifier__max_features': [3, 5, 10, 20]},
                {'classifier__n_neighbors': list(range(1,11,1)),
                  'classifier__weights': ['uniform', 'distance']}]
# Creating pipeline
pipe = Pipeline([('min_max_scaler', MinMaxScaler()),
                     ('classifier', KNeighborsClassifier())])
# Creating the grid search for the analyses using the above search space.
# I might change the scoring from accuracy to something different per instructor's
# guidance that accuracy might be the best fit due to the small number of diabetes = yes.
grid_search = GridSearchCV(pipe, search_space, cv = 5, scoring = 'accuracy', verbose = 1)
```

```
# Running the three analyses simultaneously with different settings to find the best fit for the
# training data
best_model = grid_search.fit(X_train, y_train)
```

```
from sklearn import metrics
rfc_train = best_model.predict(X_train)
print("Accuracy_Score of Training Set =", format(metrics.accuracy_score(y_train, rfc_train)))
```

```
Accuracy_Score = 1.0
```

```
An accuracy score of 1.0 suggests the model has been overfit.
```

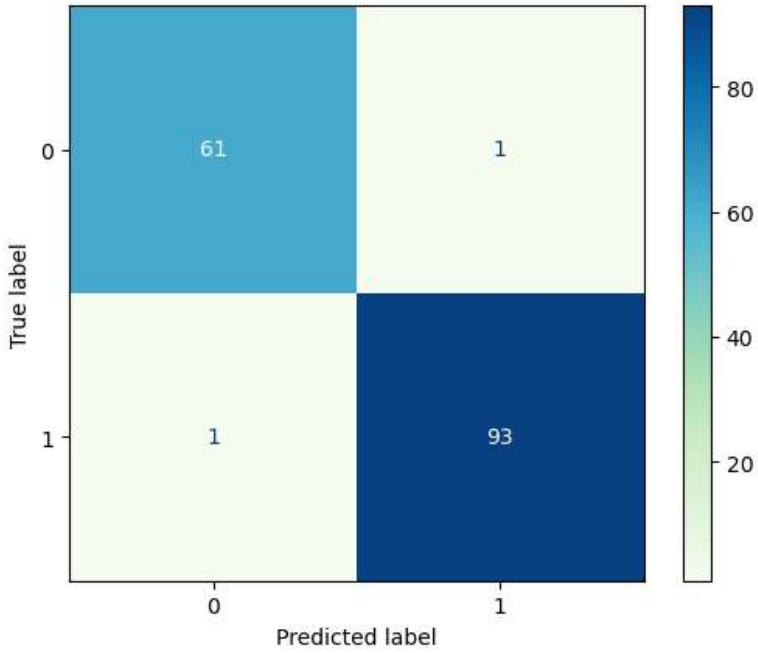
```
# Using this model with the above parameters to predict the test data outcomes (diabetes yes/no)
y_test_pred = best_model.predict(X_test)
```

```
# Finding the accuracy of the model (the y_test diabetes yes/no versus the predicted diabetes outcome)
from sklearn import metrics
print('The accuracy of the testing model is', round(metrics.accuracy_score(y_test, y_test_pred),2*100),'%')
```

```
The accuracy of the model is 0.9935897435897435 %
```

```
# Charting a confusion matrix.
from sklearn import metrics
confusion_matrix_log = metrics.plot_confusion_matrix(best_model, X_test, y_test, cmap = "GnBu")
print(confusion_matrix_log)
```

```
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay object at 0x000002A207E0CEE0>
```



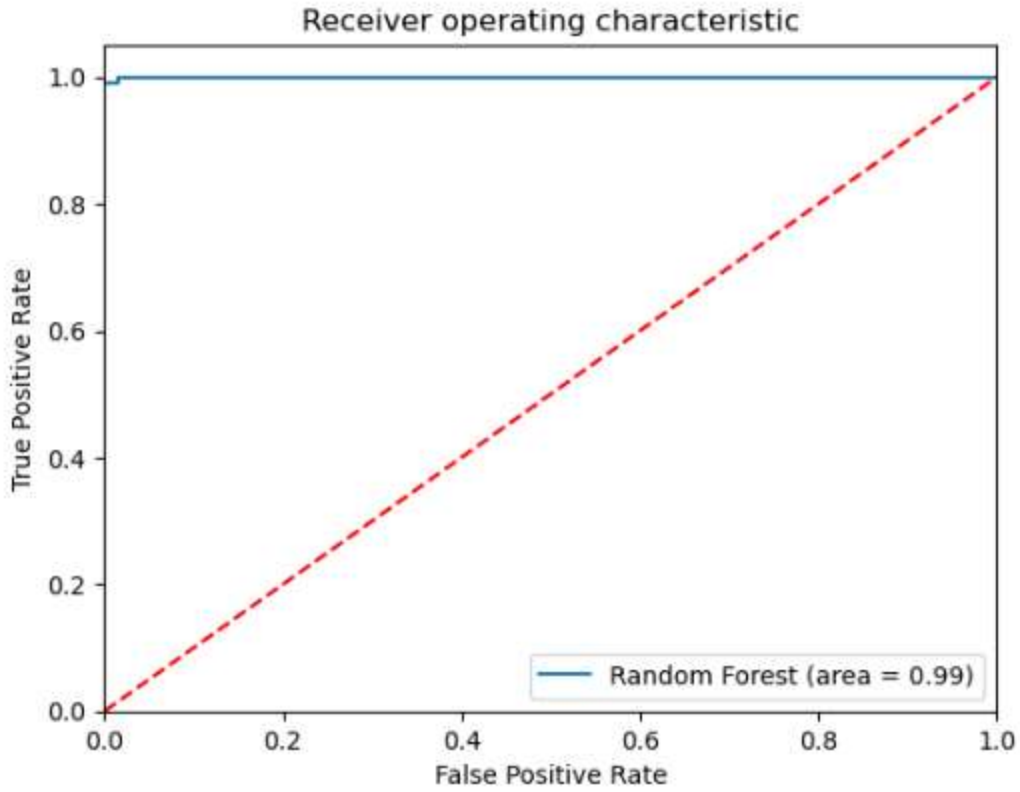

```
# Inspect the classification report
from sklearn.metrics import classification_report
print(classification_report(y_test, y_test_pred))
```

```
              precision    recall  f1-score   support

           0       0.98      0.98      0.98        62
           1       0.99      0.99      0.99        94

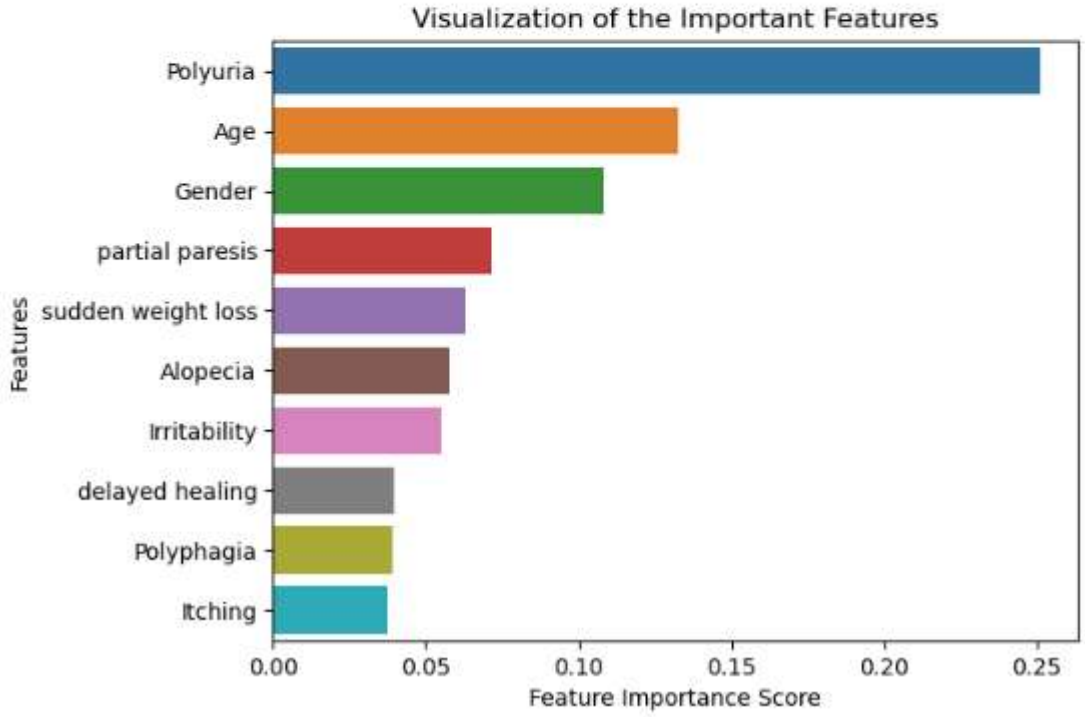
    accuracy                           0.99       156
   macro avg       0.99      0.99      0.99       156
weighted avg       0.99      0.99      0.99       156
```

```
# Using the ROC curve of the model to see the fit of the model
from sklearn.metrics import roc_auc_score
from sklearn.metrics import roc_curve
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
logit_roc_auc = roc_auc_score(y_test, best_model.predict(X_test))
fpr, tpr, thresholds = roc_curve(y_test, best_model.predict_proba(X_test)[:,1])
plt.figure()
plt.plot(fpr, tpr, label='Random Forest (area = %0.2f)' % logit_roc_auc)
plt.plot([0, 1], [0, 1],'r--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver operating characteristic')
plt.legend(loc="lower right")
plt.savefig('Log_ROC')
plt.show()
```



```
# Using feature importance to see which variables are the most important to
# this prediction model
FeatureImport = pd.Series(rf.feature_importances_, index=X_train.columns)
FeatureImport.sort_values(ascending=False, inplace=True)
# Visualize the feature importance
# Creating a bar plot
import seaborn as sns
sns.barplot(x = FeatureImport.head(10), y = FeatureImport.head(10).index)
plt.xlabel('Feature Importance Score')
plt.ylabel('Features')
plt.title("Visualization of the Important Features")
```

```
Text(0.5, 1.0, 'Visualization of the Important Features')
```



Interpretation of the Results

After running a grid search of 210 analyses of random forest classifier, KNN classifier, and logistics classifier models with multiple and varying parameters, the model of best fit was a random forest classifier model with a max of three features and 1,000 estimations. Initial results from the accuracy statistics from the random forest classifier on the training model suggest it's possible the model is overfit (with the accuracy score of 1.0). This means the training model was able to accurately classify each diabetic and non-diabetic correctly. When we fit this model to the testing model, the

accuracy decreased slightly to 99.4%. Given the accuracy of the second dataset, it's less likely the model is overfit as the accuracy should be much poorer on the testing set.

The confusion matrix above shows that the random forest classifier fit to the testing data was able to accurately predict 93 of 94 diabetics and 61 of 62 non-diabetics. Other important model statistics such as precision (the ability to not label a diabetic if they are indeed a non-diabetic), recall (the ability of the model to find all diabetics), and the f1 score (the mean of both precision and recall) are all in the 98-99% range. The receiver operating characteristic or (ROC) curve is another common tool to test the goodness of fit for a model with a binary classifier such as diabetes. The dotted line represents the ROC curve of anything appearing at random (slope=1 which makes sense). A good classifier model will stay as far away from that line as possible and head towards the top left corner (which makes sense since you are trying to max true positive rates). This model does a great job accurately predicting diabetes vs non-diabetics.

Initial Conclusion and Recommendation

My goal was to set out to find a good way to diagnose/predict which people in the population have diabetes due to measurable characteristics so as to treat them long before the disease has done a great deal of damage on their body along with costing health insurance plans and the individual themselves countless dollars on medicines, other treatment, and the long term toll of other complications due to diabetes. The model accuracy statistics are very encouraging, mostly in the 98-99% range, meaning not only can this model accurately predict diabetes due to certain symptoms (something other models that I've worked on have easily done), but it can also accurately assess those without diabetes as such (something I've struggled to build models that can accurately do).

Upon further analysis as to what features or variables were able to strengthen this model's accuracy, a feature importance graph was created to assess these top 10 features. At the top of the list

(and rightfully so due to its high correlation with diabetes) is polyuria (excessive need to urinate and/or urinate excessive amounts), followed by age, gender, and partial paresis (partial weakening of a group of muscles such as eye muscles in diabetics or diabetic neuropathy). Other than age, these top four variables are highly correlated with diabetes and give medical professionals, as well as the general public, a good starting point as far as what to look for if a person is experiencing some of these symptoms in conjunction with sudden weight loss, alopecia, irritability, or delayed healing, especially in the extremities. Given the model's high level of accuracy, no further hyperparameter tuning or other tweaking to the model or searching for better models was needed.

Milestone 3

Data versus Expectations

The "Early-Stage Diabetes Risk Prediction Dataset" was collected from a questionnaire given to patients in Sylhet, Bangladesh at the Sylhet Diabetes Hospital (UCI Machine Learning Repository, 2020). The data is complete with no missing values. Additionally, all the variables are binary (Yes/No), with Gender (also binary) and age, ranging from 16 to 90; therefore, no dummy variables will need to be created. There are no outliers to handle or any formatting issues, other than to recode the Yes/No variables (and gender) to 0 or 1. I believe that I have a good dataset to make predictions about diabetes based on the variables selected/surveyed.

Motivation

Diabetes is a health care crisis in our country. As people live longer, eat more, and exercise less, the prevalence of diabetes is increasing (www.cdc.gov). Roughly 11.3% of the population in America has diabetes, which equates to nearly 29 million people (www.cdc.gov). Even more staggering is that it is estimated that another 8.5 million persons are suffering from diabetes but have gone undiagnosed (www.cdc.gov). Furthermore, nearly 100 million adults are prediabetic, meaning if they don't make steps to change certain modifiable behaviors such as eating a low-carb diet or exercising more, they will cross into a diabetic diagnosis (American Diabetes Association, 2023). Type 2 diabetes has a latter onset and is more likely to be due to poor lifestyle choices or other health risk factors (when I mention diabetics in this paper, I'm referring to Type 2 diabetics, not the less common Type 1 diabetics that are usually diagnosed much earlier and cannot be reversed).

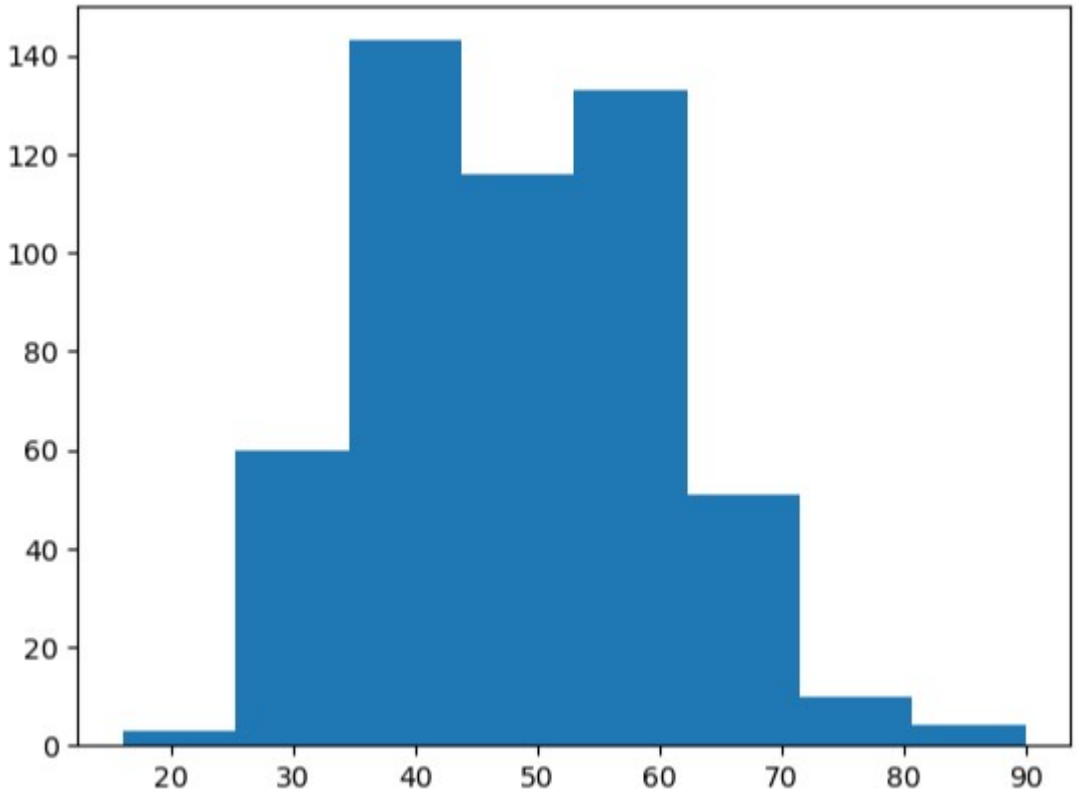
These are the groups I'd like to target with my analysis: the undiagnosed and prediabetic population, both which can benefit from making immediate life changes to either avoid diabetes or possibility even reverse diabetes (Riddle, et al., 2021). But how to target those who are untested or

aren't registering as pre-diabetic, and why is this so important? Studies show that early intervention for diabetics is important in reversing Type 2 diabetes (ADA, 2023). Diabetes, however, doesn't just affect those on the path to a diabetic diagnosis; it affects all Americans. Diabetes is a costly disease than raises health care premiums for all Americans as the cost of the maintenance diabetic drugs (either Ozempic or something similar or insulin) is mind boggling (ADA, 2023). Diagnosed diabetics tend to incur nearly $17,000 medical expenses every year compared to roughly $9,600 of non-diabetics (ADA, 2023). Furthermore, the costs of medications such as insulin continues to increase so that the use of a few vials rivals the cost of a car (ADA, 2023). Twelve months of Ozempic out of pocket isn't much better in price, costing roughly $2500 a month (www.cvs.com). In fact, the economic impact of diabetes on our society is estimated to be roughly $327 billion and is continually rising (ADA, 2023). So predicting who is going to get diabetes and heading them off through a low-carb diet as well as encouraging weight loss and exercise can really help the health of these Americans and the healthcare costs to all.

Visualizations

Histograms and boxplots will not be needed for any variable other than age to see if it's normally distributed. Here is the histogram of age showing it's mostly normally distributed:

Distribution by Age



After the model has been built, a confusion matrix as well as other visualizations (such as the area under the ROC curve, precision, recall, etc.) will be used based on what analyses is found to best fit the data (KNN, logistic regression or random forest).

Model versus Expectations

The dataset has 520 respondents which is sufficient to split the data into a training and a test data set to first train the model and then to test the model. I plan to evaluate the model selected from the series of analyses above (either random forest, KNN classification or logistic regression, up to 210 fits total) based on what works best for that type of analyses but in general evaluating the confusion matrix (especially the diabetics accurately predicting and those diabetics that weren't correctly assumed

to be diabetics by the model as that is the focus of my project), the precision, recall, f1 score, ROC (receiver operating characteristic) curve, and possibility more. I believe my expectations are likely to work; whether the model built will have an exceptional predictive power when positively identifying undiagnosed diabetics. Furthermore, whether these results of an Indian population (especially those hospitalized for already not feeling well) can be used in an American population remains to be seen. The prevalence of diabetes appears to differ by race; however, possible racial disparity in symptoms or the onset don't appear among current literature.

Milestone 2

Introduction

Diabetes is a health care crisis in our country. As people live longer, eat more, and exercise less, the prevalence of diabetes is increasing (www.cdc.gov). Roughly 11.3% of the population in America has diabetes, which equates to nearly 29 million people (www.cdc.gov). Even more staggering is that it is estimated that another 8.5 million persons are suffering from diabetes but have gone undiagnosed (www.cdc.gov). Furthermore, nearly 100 million adults are prediabetic, meaning if they don't make steps to change certain modifiable behaviors such as eating a low-carb diet or exercising more, they will cross into a diabetic diagnosis (American Diabetes Association, 2023). Type 2 diabetes has a latter onset and is more likely to be due to poor lifestyle choices or other health risk factors (when I mention diabetics in this paper, I'm referring to Type 2 diabetics, not the less common Type 1 diabetics that are usually diagnosed much earlier and cannot be reversed).

These are the groups I'd like to target with my analysis: the undiagnosed and prediabetic population, both which can benefit from making immediate life changes to either avoid diabetes or possibility even reverse diabetes (Riddle, et al., 2021). But how to target those who are untested or aren't registering as pre-diabetic, and why is this so important? Studies show that early intervention for diabetics is important in reversing Type 2 diabetes (ADA, 2023). Diabetes, however, doesn't just affect those on the path to a diabetic diagnosis; it affects all Americans. Diabetes is a costly disease than raises health care premiums for all Americans as the cost of the maintenance diabetic drugs (either Ozempic or something similar or insulin) is mind boggling (ADA, 2023). Diagnosed diabetics tend to incur nearly $17,000 medical expenses every year compared to roughly $9,600 of non-diabetics (ADA, 2023). Furthermore, the costs of medications such as insulin continues to increase so that the use of a few vials rivals the cost of a car (ADA, 2023). Twelve months of Ozempic out of pocket isn't much better in price, costing roughly $2500 a month (www.cvs.com). In fact, the economic impact of diabetes on our society

is estimated to be roughly $327 billion and is continually rising (ADA, 2023). So predicting who is going to get diabetes and heading them off through a low-carb diet as well as encouraging weight loss and exercise can really help the health of these Americans and the healthcare costs to all.

Proposal

I plan on using the "Early Stage Diabetes Risk Prediction Dataset" collected from a questionnaire given to patients in Sylhet, Bangladesh at the Sylhet Diabetes Hospital (UCI Machine Learning Repository, 2020). This dataset focuses on patients who are in the early stages of diabetes or who aren't yet diabetic (both of which I'm interested in focusing on). I found other datasets at the CDC or in Kaggle but most aren't focused on what I'm interested in predicting.

I plan on using the CRISP-DM method to tackle this project. The first step is preparing the data. In this step, I will do exploratory data analysis on the variables in the dataset (most of this has been done already) to see what my data entails. I'll handle any missing data by either deleting row-wise or substituting with the median/mode. I will also remove any outliers identified at this time in the same manner. I'll also get the data ready for analysis through feature selection (which features are correlated with diabetes and other factors as well) and dummy columns for any categorical variables. The next step is training the model. At this time, I plan to use a pipeline and a grid search to train and select the best model from a series of random forests, knn classification and logistic regression to see which model best fits the data. Depending on the prevalence of diabetics in the sample file, I might need to use a SMOTE adjustment so that my datafile isn't heavily non-diabetics. The third step is model evaluation in which generally one partakes in parameter tuning, evaluating model size and more. The fourth step is model evaluation. I plan to evaluate the model selected from the series of analyses above (either random forest, knn classification or logistic regression, up to 210 fits total) based on what works best for that type of analyses but in general evaluating the confusion matrix (especially the diabetics accurately

predicting and those diabetics that weren't correctly assumed to be diabetics by the model as that is the focus of my project), the precision, recall, f1 score, ROC (receiver operating characteristic) curve, and possibility more. The final step is deployment and that is out of the scope of this assignment (other than the final written report).

From this project, I hope to learn what factors can clue medical professionals (or even the general public) as to what risk factors to look for when assessing whether you need to be tested for diabetes (similar to the way we have infographics dealing with how to recognize you are having a stroke or breast cancer). While currently we know some of the signs of full-blown diabetes, I don't think the warning signs of pre-diabetes or even those on the path to diabetes someday is as widely known, even by medical professionals.

There are a few risks or ethical implications of using existing healthcare data for machine learning, such as identifying health disparities (Chen, et al., 2021). Another ethical issue involves the issue of healthcare data in general. While this data isn't protected by HIPAA for numerous reasons, it still deserves the privacy and respect given that these are real people suffering from real illnesses, whether it is diabetes or some other health issue they sought treatment for. Another risk is that the data is misinterpreted or a model is fit that doesn't describe the reality of diabetes and used to faultily assess someone not having diabetes when they indeed do. An opposite diagnosis (of having diabetes when they likely will not) doesn't necessarily carry as much risk given that the "treatment" would be to live a healthier lifestyle, which isn't as much harm, but it could cause undue stress and worry for that person.

And finally, if my original plan doesn't work out, as often happens, I will seek help and guidance from my professor and peers and likely start over, or return to the last step of the CRISP-DM process that I know was a success. It's possible I'll have to pick out another data set if this one isn't workable at

all and I already have one as a backup. If none of the models fit well or the features I selected aren't working well and the model isn't a good fit at all, I'll back up a few steps and assess what the issue is and dedicate extra time to work towards a successful finish.

References

American Diabetes Association. (2023). Diabetes Statistics. Retrieved from www.diabetes.org.

Chen, Irene, Emma Pierson, Sherri Rose, Shalmali Joshi, Kadija Ferryman, Marzyeh Ghassemi. Ethical Machine Learning in Healthcare. Annual Review of Biomedical Data Science 2021 4:1, 123-144

Centers for Disease Control and Prevention. National Diabetes Statistics Report website. https://www.cdc.gov/diabetes/data/statistics-report/index.html. Accessed March 2023.

Economic costs of diabetes in the US in 2017. Diabetesjournals.org. (n.d.). Retrieved March 26, 2023, from https://diabetesjournals.org/care/article/41/5/917/36518/Economic-Costs-of-Diabetes-in-the-U-S-in-2017

Riddle, Matthew, et. Al. Consensus Report: Definition and Interpretation of Remission in Type 2 Diabetes. *Diabetes Care* 2021;44(10):2438–2444

UCI Machine Learning Repository. (2020, July 12). Early-stage diabetes risk prediction dataset. https://archive.ics.uci.edu/ml/datasets/Early+stage+diabetes+risk+prediction+dataset.